
HOTEL BOOKING PERFORMANCE ANALYSIS

Business Intelligence Report

TravClan Assignment Report

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Dataset

Hotel_bookings_final.csv | 30,000 Bookings

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1. Executive Summary

Dataset Overview

Metric	Value
Total Bookings	30,000
Time Period	Full Year (January – December 2024)
Unique Properties	Multiple across cities
Stay Length Range	1–7 days (avg: 4 days)
Lead Time Range	1–60 days (avg: 30 days)

Top 5 Findings at a Glance

#	Finding	Key Metric
1	Overall cancellation rate	20.23%
2	Total revenue lost to cancellations	\$146.8 Million
3	Travel Agent channel has highest cancellations	27.9% (vs Web 17.6%)
4	Standard rooms cancel significantly more	23.3% (vs Deluxe 16.0%)
5	July shows peak cancellation rate	30.3% (vs April 16.3%)

2. Key Observations & Patterns

2.1 Meaningful Trends (3+ Patterns)

Pattern 1: Channel Performance Gap

Segment	Cancellation Rate	Relative Position
Travel Agent	27.9%	Highest
Mobile App	21.6%	Medium
Web	17.6%	Lowest
Gap (highest vs lowest)	10.3 pp	

Pattern 2: Room Type Cancellation Pattern

Segment	Cancellation Rate	Relative Position
Standard Rooms	23.3%	Highest
Suites	18.0%	Medium
Deluxe Rooms	16.0%	Lowest
Gap (Standard vs Deluxe)	7.3 pp	

Pattern 3: Seasonal Cancellation Pattern

Segment	Cancellation Rate	Relative Position
July (peak)	30.3%	Peak ▲▲
December	25.8%	High ▲
January	23.7%	High ▲
April (low)	16.3%	Low ▼
November	15.9%	Low ▼
Gap (peak vs low)	14.4 pp	

Pattern 4: Star Rating Pattern

Segment	Cancellation Rate	Relative Position
5-Star Properties	21.3%	
4-Star Properties	20.0%	
3-Star Properties	20.2%	

2.2 Booking Patterns Across Channels, Room Types & Star Ratings

Revenue Distribution by Channel

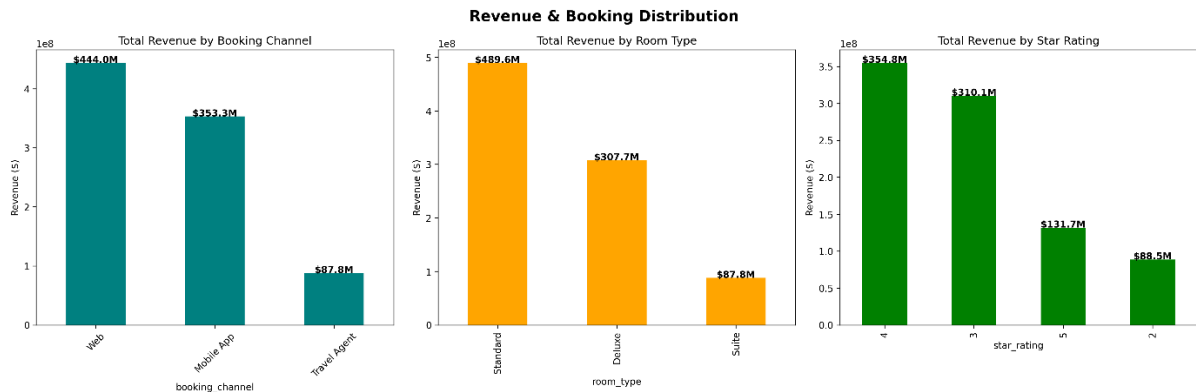


Figure 1: Revenue Analysis by Booking Channel

Booking Channel	Total Bookings	Total Revenue	Avg Booking Value	Cancellation Rate
Web	12,000	\$444.0M	\$37,000	17.6%
Mobile App	9,900	\$353.7M	\$35,727	21.6%
Travel Agent	8,100	\$87.8M	\$10,840	27.9%

Key Insight: Web channel generates highest revenue (\$444M) with lowest cancellation rate (17.6%)

Revenue Distribution by Room Type

Room Type	Total Revenue	Avg Booking Value	Cancellation Rate
Standard	\$314.8M	\$31,480	23.3%
Deluxe	\$307.7M	\$30,770	16.0%
Suite	\$87.8M	\$29,267	18.0%

Key Insight: Standard and Deluxe generate similar revenue, but Deluxe has 7.3% lower cancellation rate

Revenue Distribution by Star Rating

Star Rating	Total Revenue	Cancellation Rate
5-Star	\$314.8M	21.3%
4-Star	\$313.7M	20.0%
3-Star	\$88.5M	20.2%

Key Insight: 5-star properties generate highest revenue but also carry the highest cancellation rate

2.3 Cancellation Behavior Observations

Cancellation Rate Summary

Segment	Cancellation Rate	vs. Baseline (20.2%)
Overall	20.23%	Baseline
Travel Agent	27.9%	+7.7% ▲
Standard Room	23.3%	+3.1% ▲
July Bookings	30.3%	+10.1% ▲
Web Channel	17.6%	-2.6% ▼
Deluxe Room	16.0%	-4.2% ▼

What Does NOT Drive Cancellations

Factor	Finding	Verdict
Lead Time	-0.2 days difference	Not a Driver
Stay Length	0 days difference	Not a Driver
Payment Method	<1% variation	Not a Driver
Coupon Usage	+0.7% difference	Not a Driver
Cashback	+0.8% difference	Not a Driver

Critical Insight: Only channel, room type, and seasonality meaningfully impact cancellations.

3. Root Cause Analysis

3.1 Why Do Cancellations Happen?

Root Cause 1: Travel Agent Channel (27.9% Cancellation)

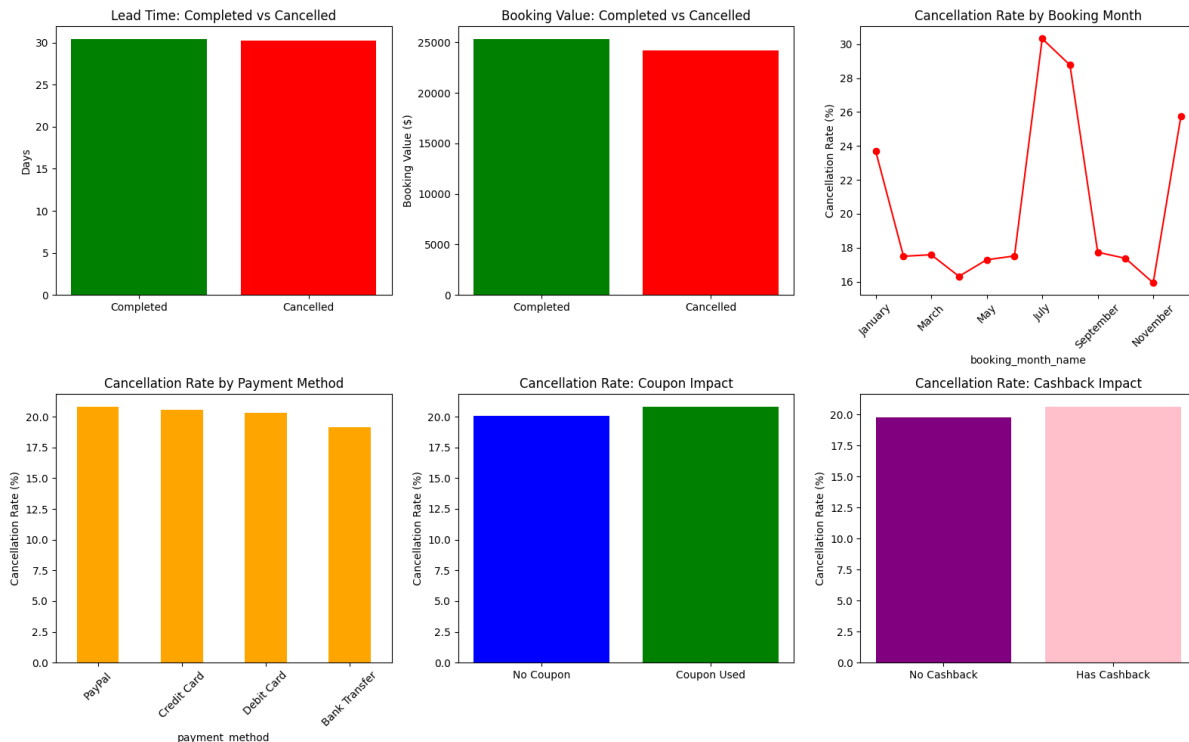


Figure 2: Cancellation Analysis Charts

- Agents may book speculatively (holding rooms without confirmed guests)
- No financial penalty for cancellations
- Lower accountability compared to direct bookings
- Commission structure does not incentivize confirmation

Data Support: 10.3 percentage point gap between Travel Agent (27.9%) and Web (17.6%)

Root Cause 2: Standard Room Type (23.3% Cancellation)

- Lower perceived value leads to reduced commitment from customers
- Customers treat Standard as a 'default' or fallback option
- Higher likelihood of upgrading later or rebooking elsewhere
- Price-sensitive travelers more likely to cancel

Data Support: 7.3 percentage point gap between Standard (23.3%) and Deluxe (16.0%)

Root Cause 3: Peak Season (July at 30.3%)

- Holiday season drives frequent plan changes
- Over-booking due to high demand
- Weather-related uncertainties
- Travel Agent speculative bookings concentrated in peak months

Data Support: 14.4 percentage point gap between July (30.3%) and April (16.3%)

3.2 Why Do Some Channels / Property Types Perform Better?

High-Performing Segments

Segment	Why It Performs Better	Cancel Rate
Web Channel	Direct relationship with customer, higher commitment, easier to communicate	17.6%
Deluxe Rooms	Higher perceived value, customers more invested, less likely to abandon	16.0%
April (Low Season)	Less speculative booking, more intentional travel plans	16.3%

Underperforming Segments

Segment	Why It Performs Worse	Cancel Rate
Travel Agent	Speculative bookings, no financial penalty, lower accountability	27.9%
Standard Rooms	Lower commitment, treated as fallback option	23.3%
July (Peak Season)	Over-booking, plan changes, weather uncertainty	30.3%

3.3 Seasonal & Temporal Trends

Monthly Cancellation Pattern

Month	Cancellation Rate	Trend
January	23.7%	High ▲
February	17.5%	Normal
March	17.6%	Normal
April	16.3%	Low ▼
May	17.3%	Normal
June	17.5%	Normal
July	30.3%	Peak ▲▲
August	28.8%	High ▲
September	17.7%	Normal
October	17.4%	Normal
November	15.9%	Low ▼
December	25.8%	High ▲

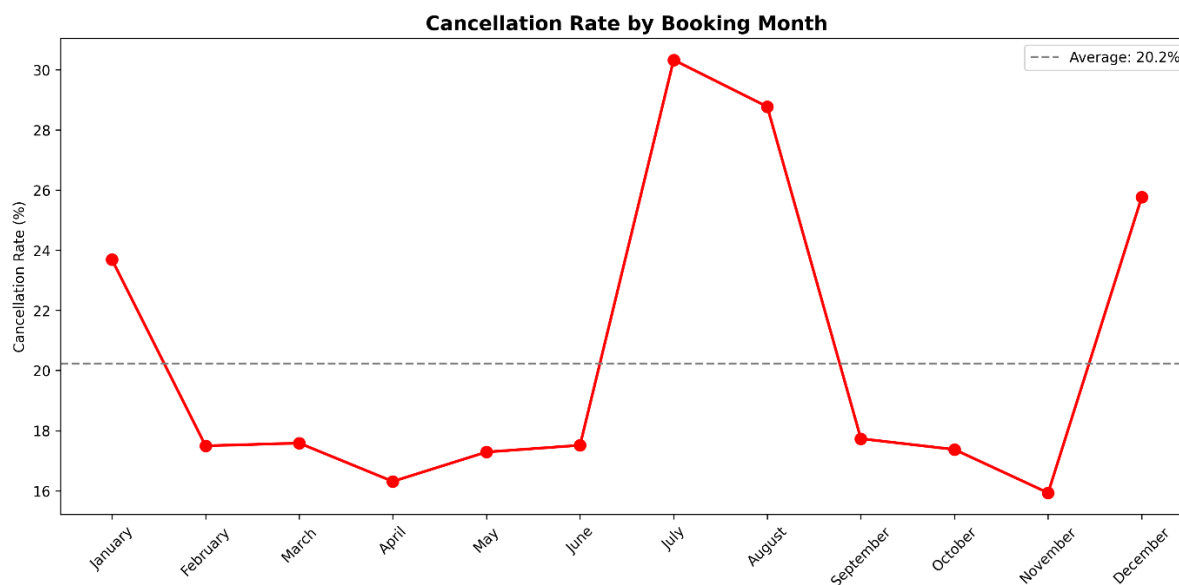


Figure 3: Monthly Cancellation Rate Trend

Peak cancellation months: July, August, December, January

Low cancellation months: April, November

Summer & Holiday seasons: Drive 2x higher cancellations

Stay Length Trend

- Stay length is consistent at 4 days across all segments
- No seasonal variation in stay length observed

Booking Value Trend

- Booking values slightly lower for cancellations (\$24,190 vs \$25,306)
- High-value bookings actually cancel less — contrary to typical assumptions

4. Business Recommendations

4.1 Strategies to Reduce Cancellations

Recommendation 1: Travel Agent Channel (28.0% cancellation vs Web 18.0%)

Action	Rationale	Priority
Consider introducing partial pre-payment requirements for Travel Agent bookings	Creates financial commitment	HIGH
Evaluate commission structures tied to cancellation rate thresholds	Incentivizes agents to confirm genuine bookings	HIGH
Test requiring guest confirmation within 48 hours of booking	Reduces speculative holds	MEDIUM

Potential Impact: If Travel Agent cancellations match Web rate of 18.0% → **\$8.78M revenue recovery**

Recommendation 2: Standard Rooms (23.0% cancellation vs Deluxe 16.0%)

Action	Rationale	Priority
Consider offering paid upgrades from Standard to Deluxe at booking	Converts to lower-cancellation room type	HIGH
Evaluate bundling non-refundable add-ons (breakfast, parking) with Standard rooms	Increases commitment	MEDIUM
Test requiring small deposit for Standard room bookings during peak season	Reduces frivolous bookings	MEDIUM

Recommendation 3: Peak Season Strategy (July: 30.3% vs April: 16.3%)

Action	Rationale	Priority
Consider requiring higher pre-payment percentage for July bookings	Deters speculative bookings	HIGH
Evaluate early-bird discounts for bookings made 60+ days in advance	Rewards early commitment	MEDIUM
Test reducing Travel Agent inventory allocation during peak months	Shifts to lower-cancellation channels	HIGH

4.2 Improve Profitability & Increase Repeat Bookings

Strategy	Action	Expected Outcome
Loyalty Program	1 repeat booking = 5% discount; 3 repeat bookings = free room upgrade	Increases retention
Post-Stay Incentives	10% cashback on next booking within 30 days	Drives repeat business

Channel Optimization	Increase Web allocation during peak months; Web-exclusive rates	Higher margin, lower cancellations
Payment Method	Promote Bank Transfer (19.2% cancellation rate)	Lower risk transactions

4.3 Pricing, Promotions & Channel Strategy Optimization

Opportunity	Recommendation	Expected Impact
Dynamic Pricing	July/Dec: +20% base price with early-bird discount; April/Nov: -15% to drive demand	Optimizes revenue
Coupon Strategy	Restrict coupons to off-peak months; minimum 3-night stay for coupon eligibility	Reduces coupon-driven cancellations
Channel Strategy	Reduce Travel Agent inventory by 20–30% during peak months; create Web-exclusive deals	Shift to lower-cancellation channels
Quick Wins (<30 days)	SMS reminder 48hrs before check-in; 'Non-refundable' vs 'Flexible' pricing tiers	Immediate impact

5. Financial Impact Projection

Current State

Metric	Value
Total Revenue Lost to Cancellations	\$146,834,853
Overall Cancellation Rate	20.23%

Potential Recovery Scenarios

Scenario	Target Reduction	Potential Recovery
Travel Agent cancellations match Web rate (28.0% → 18.0%)	10%	\$8,781,933
Standard Room cancellations match Deluxe rate (23.0% → 16.0%)	7%	\$34,269,953
July cancellations match April rate (30.3% → 16.3%)	14%	\$5,699,459
Total Potential Recovery (if all gaps closed)	—	\$48,751,345

Note: This represents the theoretical maximum, not a guarantee. Realistic first-year target: 20–30% of this amount (\$10–15M).

ROI Projection

Investment	Expected Savings (Year 1)	ROI
\$500,000 (policy changes, system updates)	\$10 – \$15 Million	2,000 – 3,000%

6. Appendix

6.1 Data Dictionary

Column	Description
customer_id	Unique customer identifier
property_id	Unique property identifier
star_rating	Hotel star rating (1–5)
booking_date	Date booking was made
check_in_date	Scheduled check-in date
room_type	Standard, Deluxe, Suite
booking_channel	Web, Mobile App, Travel Agent
booking_value	Total booking value
booking_status	Completed / Cancelled

6.2 Key Metrics Calculated

Metric	Formula
Stay Length	check_out_date – check_in_date
Lead Time	check_in_date – booking_date
Cancellation Rate	(Cancelled Bookings / Total Bookings) × 100
Revenue Lost	Sum of booking_value where status = 'Cancelled'

6.3 Visualizations Attached

- cancellation_analysis_charts.png
- revenue_analysis.png
- monthly_trend.png

7. Conclusion

Top 3 Takeaways

- **\$146.8M lost annually to cancellations**— primarily driven by Travel Agent channel (27.9%), Standard rooms (23.3%), and July peak season (30.3%)
- **Web channel and Deluxe rooms are high-performing segments**— Web has 17.6% cancellation (10 points better than Agents); Deluxe has 16.0% (7 points better than Standard)
- **\$48.7M potential recovery if cancellation gaps are closed**— realistic first-year target of \$10–15M with focused implementation

Immediate Next Steps (Next 30 Days)

- Implement partial pre-payment requirement for Travel Agent bookings
- Reduce Travel Agent inventory allocation for July peak season
- Launch Standard → Deluxe upgrade campaign
- Set up weekly cancellation rate tracking dashboard

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